

Initiating Search

November 4, 2025, 10:58 AM

• Search:

CAS Lexicon Search:

Sonogashira coupling reaction - Preferred Concept

Search Tasks

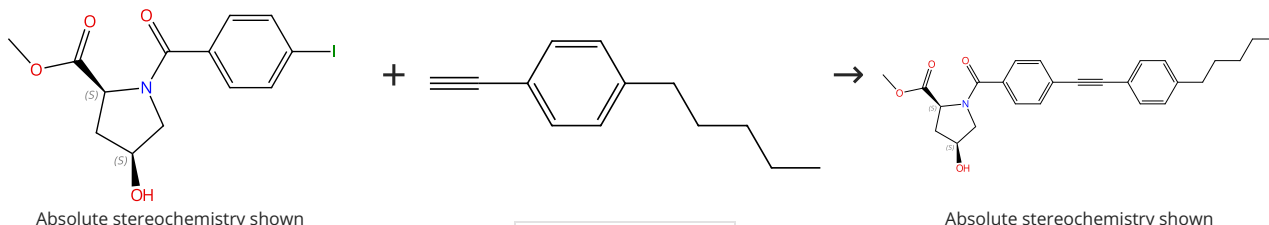
Task	Result Type	View
Returned Reference Results from CAS Lexicon (8,018)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,372)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Name:	Bioorganic & Medicinal Chemistry, European Journal of Medicinal Chemistry, Journal of Medicinal Chemistry, Medicinal Chemistry (Sharjah, United Arab Emirates), Medicinal Chemistry Research, Organic Process Research & Development, Bioorganic & Medicinal Chemistry Letters, Organic & Biomolecular Chemistry	

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (71)

31-180-CAS-16307822

Steps: 1 Yield: 100%

Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography

By: Kalinin, Dmitrii V.; et al

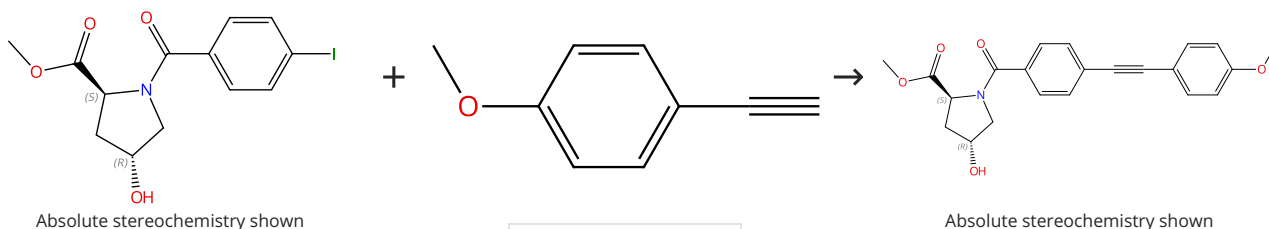
Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.

1.1 **Reagents:** Triethylamine
Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium
Solvents: Acetonitrile; 2 h, rt; overnight, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (101)

31-180-CAS-16307801

Steps: 1 Yield: 100%

Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography

By: Kalinin, Dmitrii V.; et al

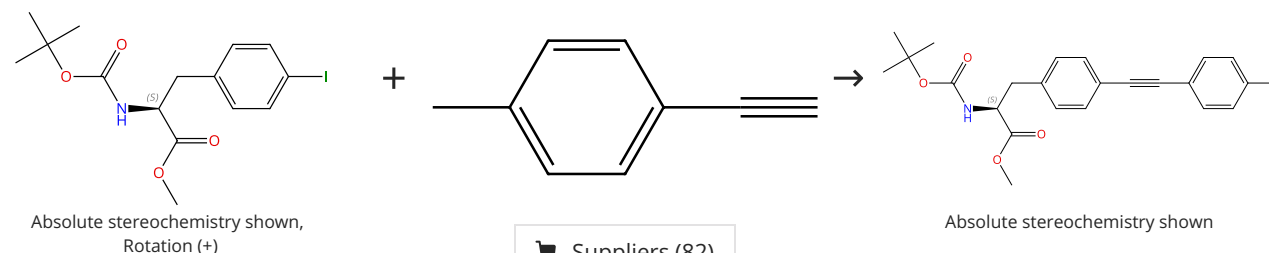
Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.

1.1 **Reagents:** Triethylamine
Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium
Solvents: Acetonitrile; 2 h, rt; overnight, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

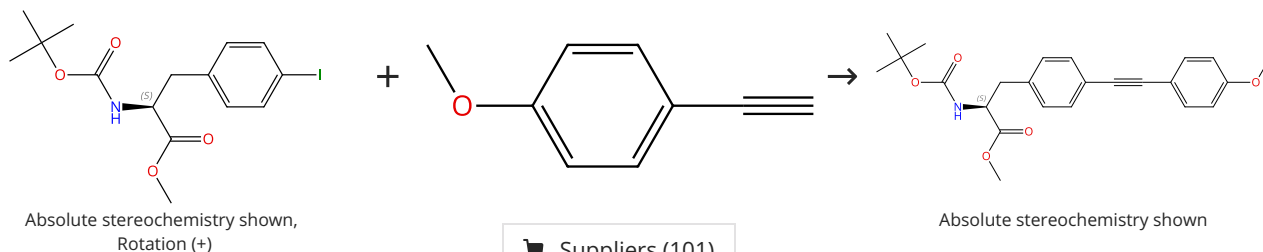

 Suppliers (82)

 Suppliers (52)

31-180-CAS-17227773 Steps: 1 Yield: 100%	A palladium and gold catalytic system enables direct access to O- and S-linked non-natural glyco-conjugates By: Jeon, Min Ho; et al Organic & Biomolecular Chemistry (2016), 14(48), 11518-11524.
1.1 Reagents: Triethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium, Gold iodide Solvents: Acetonitrile; rt; 18 h, 80 °C	
Experimental Protocols	

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



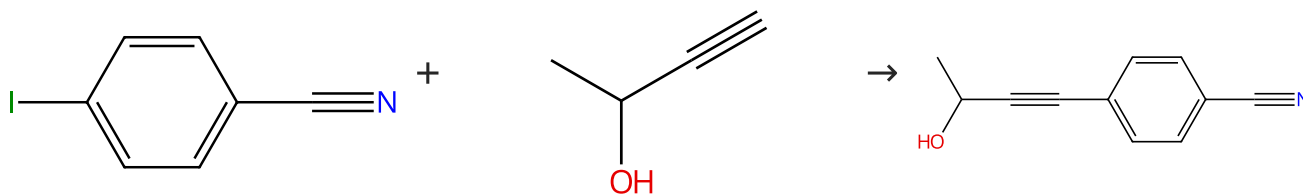
Suppliers (101)

Suppliers (52)

31-180-CAS-17227771 Steps: 1 Yield: 100%	A palladium and gold catalytic system enables direct access to O- and S-linked non-natural glyco-conjugates By: Jeon, Min Ho; et al Organic & Biomolecular Chemistry (2016), 14(48), 11518-11524.
1.1 Reagents: Triethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium, Gold iodide Solvents: Acetonitrile; rt; 18 h, 80 °C	
Experimental Protocols	

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

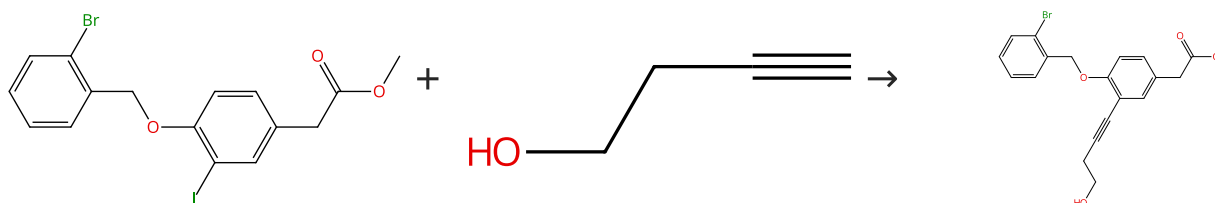
Suppliers (52)

Suppliers (5)

31-180-CAS-11098928 Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines By: Saravanan, Thangavelu; et al Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C 1.2 10 - 24 h, 85 °C	
Experimental Protocols	

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (15)

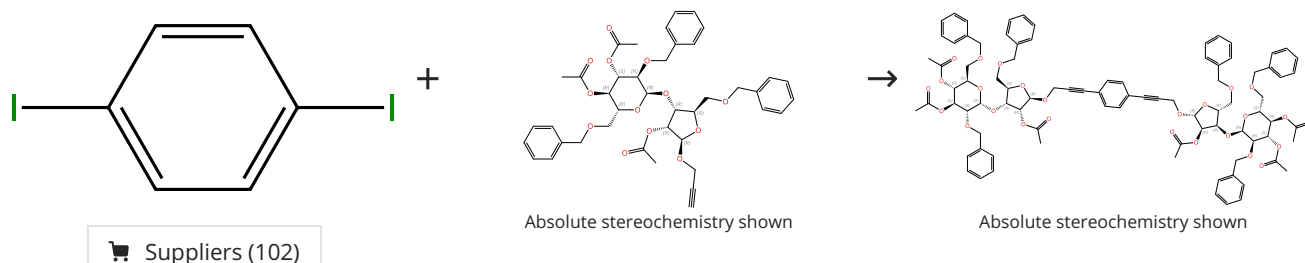
Suppliers (95)

Suppliers (11)

31-180-CAS-13336934	Steps: 1 Yield: 100%	A New Efficient Synthetic Route for the Synthesis of the Antiallergic Drug, Olopatadine Hydrochloride, via Stereospecific Palladium-Catalyzed Reaction
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Dimethylformamide; 5 h, 25 °C		By: Nishimura, Koichiro; et al
1.2 Reagents: Water		Organic Process Research & Development (2012), 16(2), 225-231.
Experimental Protocols		

Scheme 7 (1 Reaction)

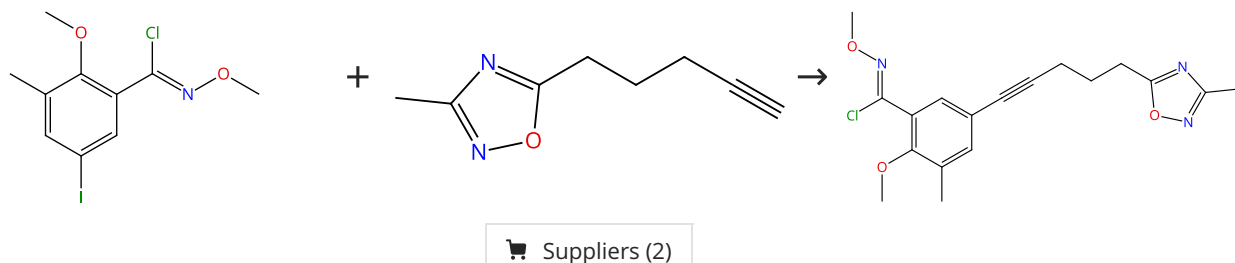
Steps: 1 Yield: 100%



31-180-CAS-2191942	Steps: 1 Yield: 100%	Synthesis of Potent Agonists of the D-myo-Inositol 1,4,5-Trisphosphate Receptor Based on Clustered Disaccharide Polyphosphate Analogues of Adenophostin A
1.1 Reagents: Triethylamine, Cuprous iodide Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide		By: De Kort, Martin; et al
1.2 Solvents: Dimethylformamide		Journal of Medicinal Chemistry (2000), 43(17), 3295-3303.
1.3 Solvents: Diethyl ether		
Experimental Protocols		

Scheme 8 (1 Reaction)

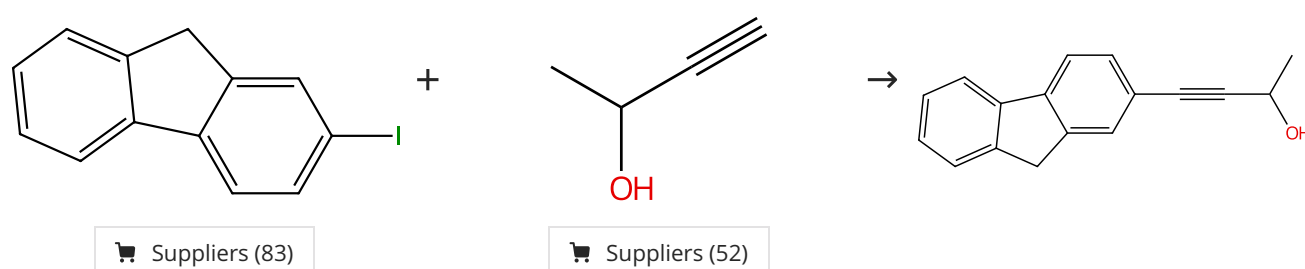
Steps: 1 Yield: 100%



31-180-CAS-15929527	Steps: 1 Yield: 100%	Systematic evaluation of methyl ester bioisosteres in the context of developing alkenyldiarylmethanes (ADAMs) as non-nucleoside reverse transcriptase inhibitors (NNRTIs) for anti-HIV-1 chemotherapy
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 0 °C; 20 h, rt		By: Hoshi, Ayako; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2016), 24(13), 3006-3022.

Scheme 9 (1 Reaction)

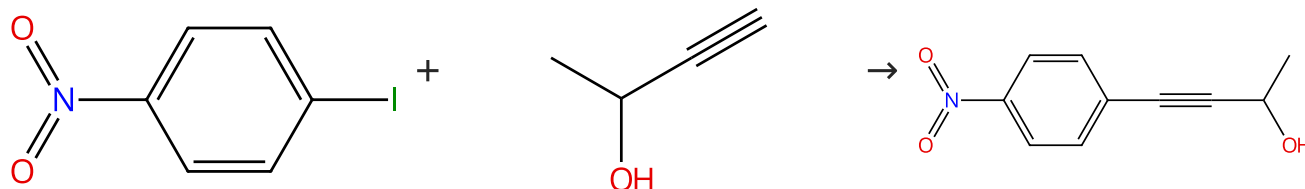
Steps: 1 Yield: 100%



31-180-CAS-4084193	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines By: Saravanan, Thangavelu; et al Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		
1.2 10 - 24 h, 85 °C		
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

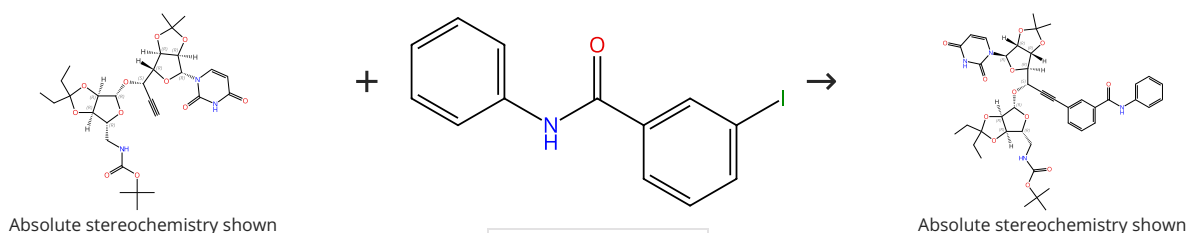
Suppliers (52)

Suppliers (9)

31-180-CAS-1724564	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines By: Saravanan, Thangavelu; et al Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		
1.2 10 - 24 h, 85 °C		
Experimental Protocols		

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

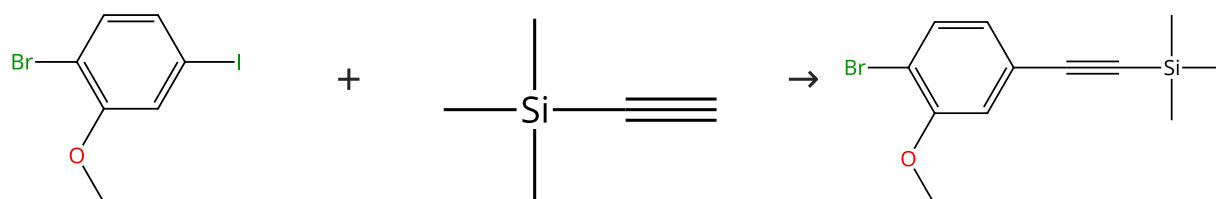
Absolute stereochemistry shown

Suppliers (14)

31-614-CAS-33440875	Steps: 1 Yield: 100%	Design, synthesis and biological evaluation of simplified analogues of MraY inhibitory natural product with rigid scaffold By: Okamoto, Kazuhiro; et al Bioorganic & Medicinal Chemistry (2022), 55, 116556.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 14 h, rt		
1.2 Reagents: Ammonium chloride Solvents: Water; rt		
Experimental Protocols		

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (66)

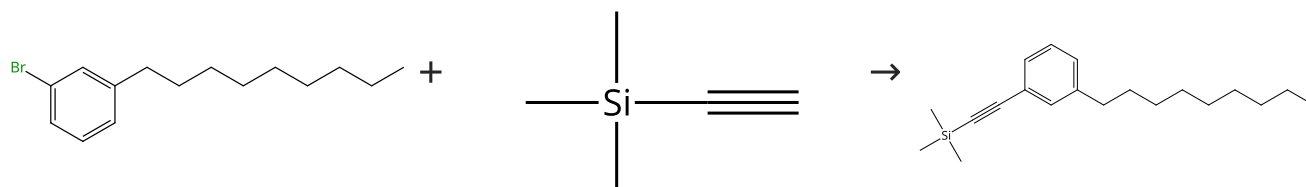
Suppliers (98)

Supplier (1)

31-180-CAS-20093966 Steps: 1 Yield: 100%	Using Nature's polyenes as templates: studies of synthetic xanthomonadin analogues and realising their potential as antioxidants By: Madden, Katrina S.; et al Organic & Biomolecular Chemistry (2019), 17(15), 3752-3759.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium; 16 h, rt	

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



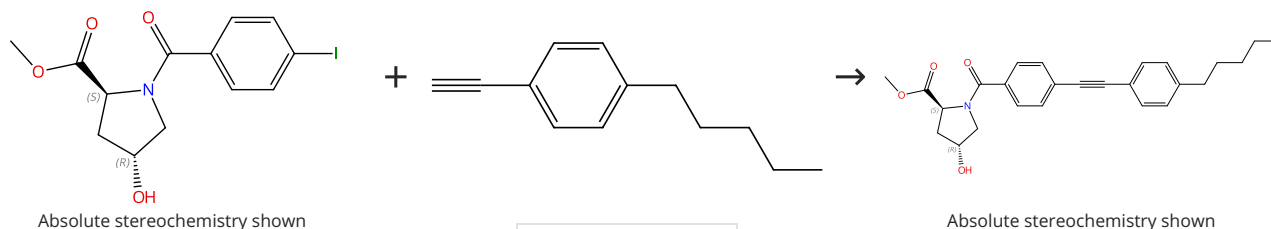
Supplier (1)

Suppliers (98)

31-614-CAS-37992505 Steps: 1 Yield: 100%	Phenyl dialkynylcarbinols, a Bioinspired Series of Synthetic Antitumor Acetylenic Lipids By: Bossuat, Margaux; et al Journal of Medicinal Chemistry (2023), 66(20), 13918-13945.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium; rt; 15 h, 80 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	
Experimental Protocols	

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Suppliers (71)

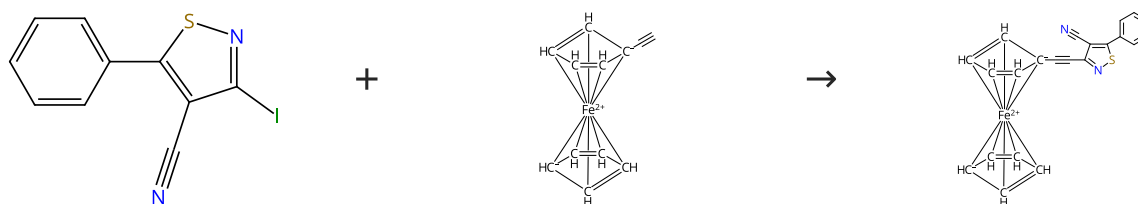
Absolute stereochemistry shown

Supplier (1)

31-180-CAS-16307802 Steps: 1 Yield: 100%	Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography By: Kalinin, Dmitrii V.; et al Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Acetonitrile; 2 h, rt; overnight, rt	
Experimental Protocols	

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



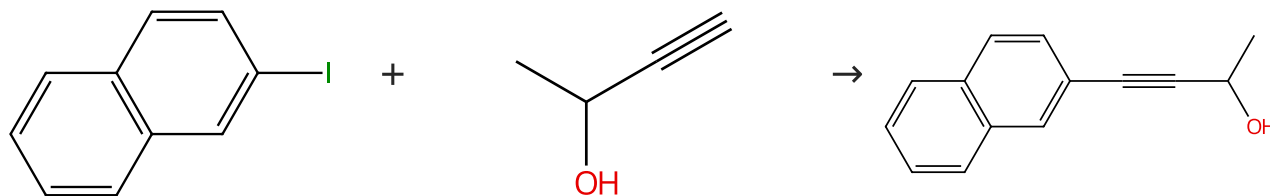
Suppliers (5)

Suppliers (66)

31-180-CAS-10842729	Steps: 1 Yield: 100%	New regiospecific isothiazole C-C coupling chemistry
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Dimethylformamide; 0.5 h, 100 °C		By: Christoforou, Irene C.; et al Organic & Biomolecular Chemistry (2006), 4(19), 3681-3693.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

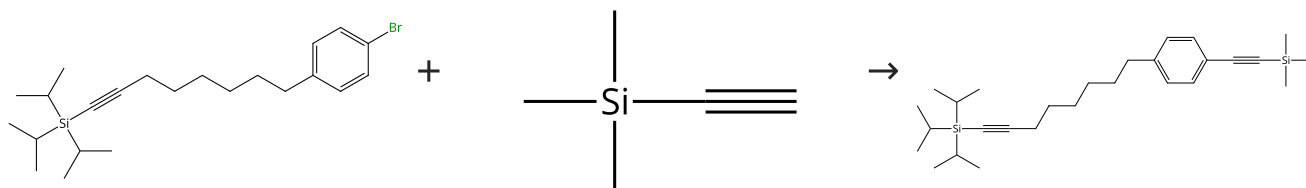
Suppliers (52)

Supplier (1)

31-180-CAS-13234432	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		By: Saravanan, Thangavelu; et al
1.2 10 - 24 h, 85 °C		Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
Experimental Protocols		

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%

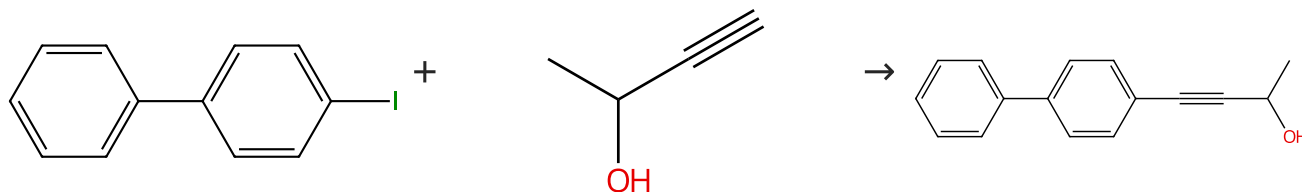


Suppliers (98)

31-614-CAS-37992628	Steps: 1 Yield: 100%	Phenyl dialkynylcarbinols, a Bioinspired Series of Synthetic Antitumor Acetylenic Lipids
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium; rt; 15 h, 80 °C		By: Bossuat, Margaux; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Journal of Medicinal Chemistry (2023), 66(20), 13918-13945.
Experimental Protocols		

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

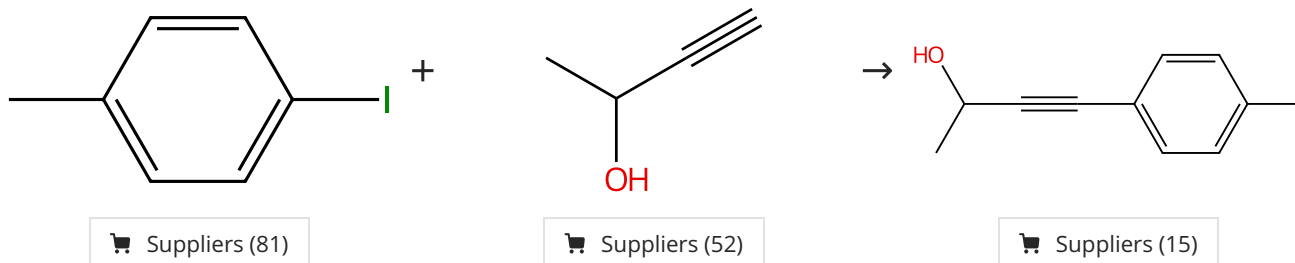
Suppliers (52)

Suppliers (7)

31-180-CAS-15362815	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines By: Saravanan, Thangavelu; et al Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		
1.2 10 - 24 h, 85 °C		
Experimental Protocols		

Scheme 19 (1 Reaction)

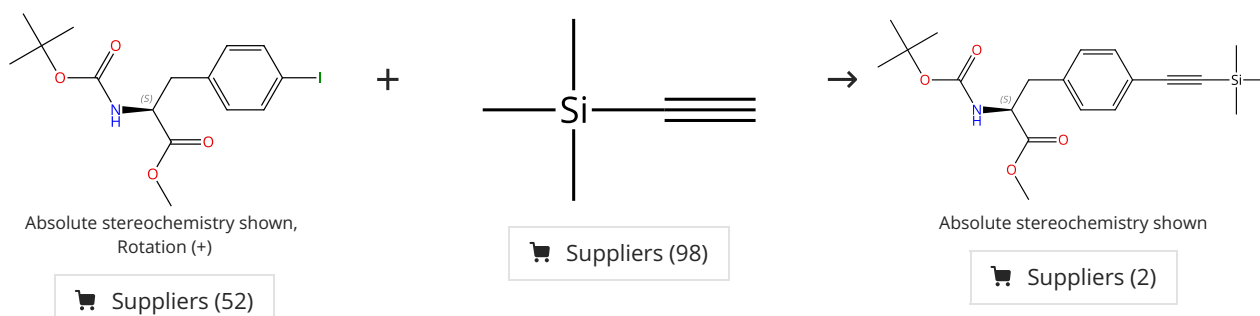
Steps: 1 Yield: 100%



31-180-CAS-8693657	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines By: Saravanan, Thangavelu; et al Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		
1.2 10 - 24 h, 85 °C		
Experimental Protocols		

Scheme 20 (1 Reaction)

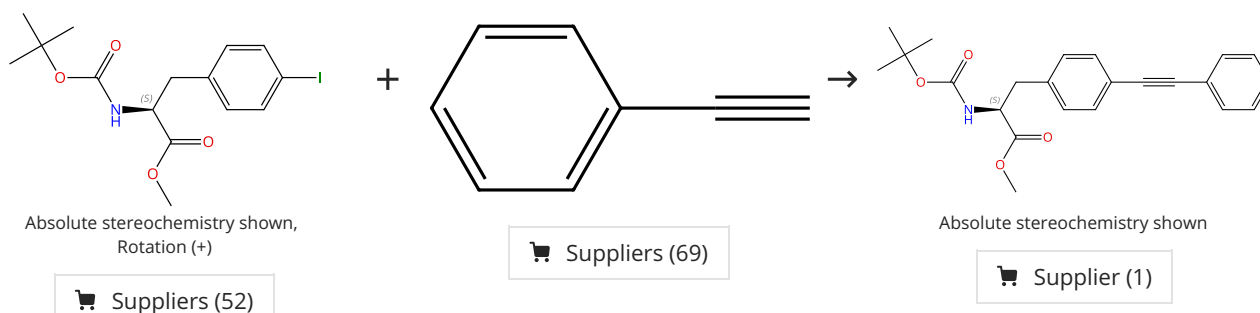
Steps: 1 Yield: 100%



31-180-CAS-17227770	Steps: 1 Yield: 100%	A palladium and gold catalytic system enables direct access to O- and S-linked non-natural glyco-conjugates By: Jeon, Min Ho; et al Organic & Biomolecular Chemistry (2016), 14(48), 11518-11524.
1.1 Reagents: Triethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium, Gold iodide Solvents: Acetonitrile; rt; 18 h, 80 °C		
1.2 Solvents: Water		
Experimental Protocols		

Scheme 21 (1 Reaction)

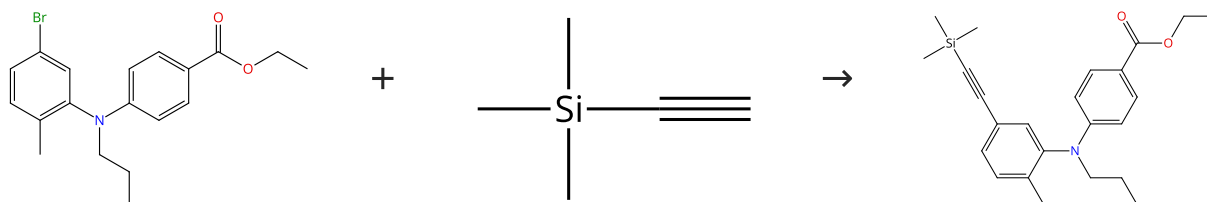
Steps: 1 Yield: 100%



31-180-CAS-1722772 Steps: 1 Yield: 100%	A palladium and gold catalytic system enables direct access to O- and S-linked non-natural glyco-conjugates By: Jeon, Min Ho; et al Organic & Biomolecular Chemistry (2016), 14(48), 11518-11524.
1.1 Reagents: Triethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium, Gold iodide Solvents: Acetonitrile; rt; 18 h, 80 °C	
Experimental Protocols	

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



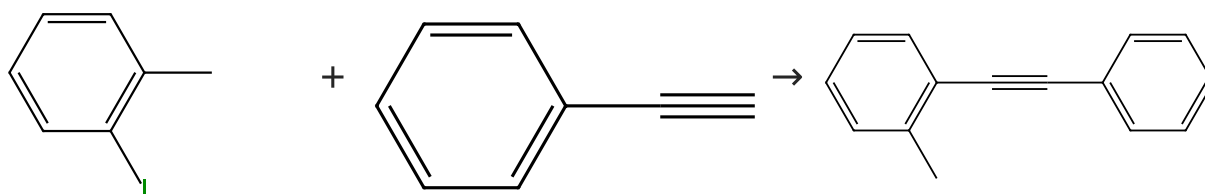
Suppliers (2)

Suppliers (98)

31-180-CAS-435484 Steps: 1 Yield: 100%	Novel retinoid X receptor (RXR) antagonists having a dicarba-closo-dodecaborane as a hydrophobic moiety By: Ohta, Kiminori; et al Bioorganic & Medicinal Chemistry Letters (2004), 14(23), 5913-5918.
1.1 Reagents: Diisopropylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran	
Experimental Protocols	

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

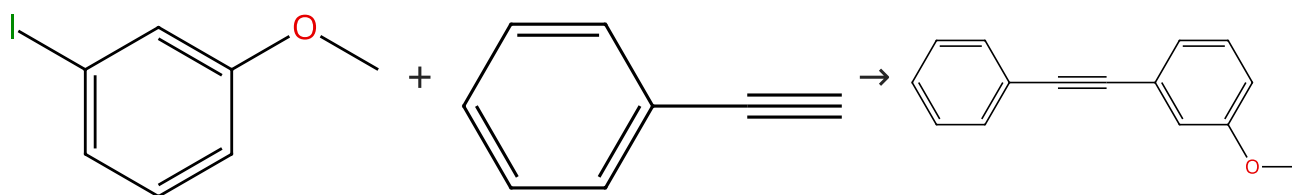
Suppliers (69)

Suppliers (31)

31-180-CAS-4574624 Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts By: Pandarus, Valerica; et al Organic Process Research & Development (2012), 16(1), 117-122.
1.1 Reagents: Potassium carbonate Catalysts: Palladium (silica-supported) Solvents: Ethanol; 4 h, reflux	
Experimental Protocols	

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

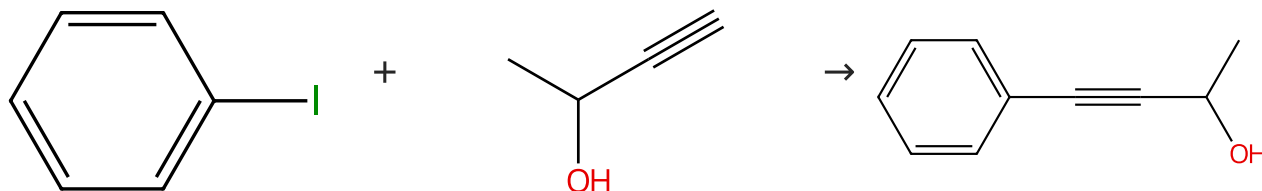
Suppliers (69)

Suppliers (21)

31-180-CAS-307163	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Potassium carbonate Catalysts: Palladium (silica-supported) Solvents: Ethanol; 4 h, reflux	By: Pandarus, Valerica; et al	
Experimental Protocols	Organic Process Research & Development (2012), 16(1), 117-122.	

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

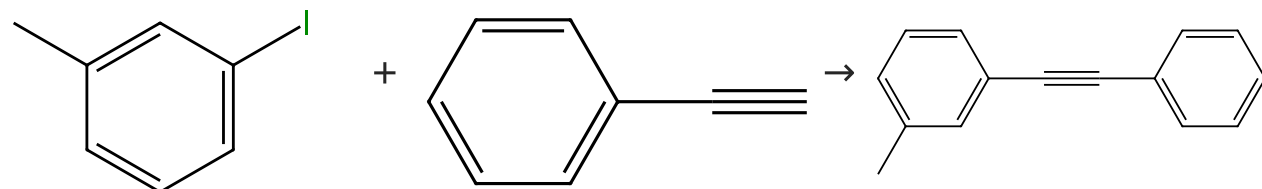
Suppliers (52)

Suppliers (46)

31-180-CAS-5297589	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C	By: Saravanan, Thangavelu; et al	
1.2 10 - 24 h, 85 °C	Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.	
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (73)

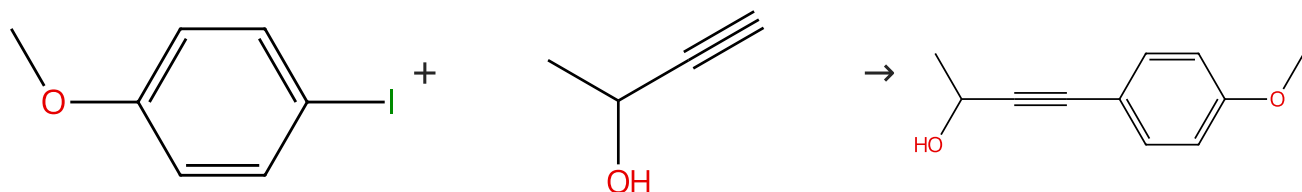
Suppliers (69)

Suppliers (19)

31-180-CAS-6697061	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Potassium carbonate Catalysts: Palladium (silica-supported) Solvents: Ethanol; 4 h, reflux	By: Pandarus, Valerica; et al	
Experimental Protocols	Organic Process Research & Development (2012), 16(1), 117-122.	

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

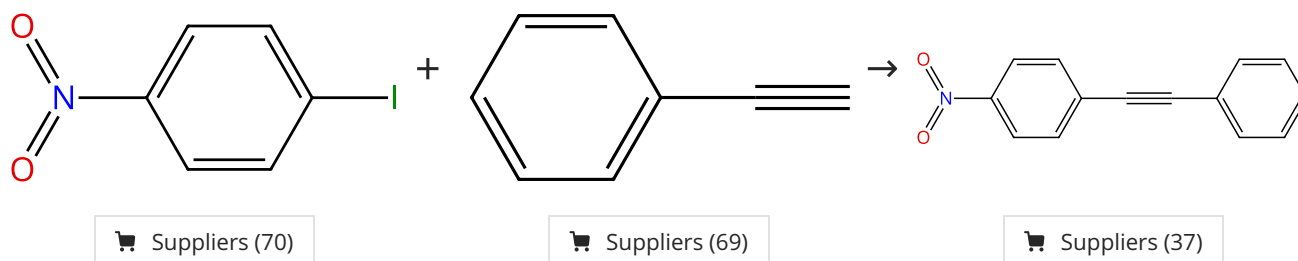
Suppliers (52)

Suppliers (14)

31-180-CAS-15082137	Steps: 1 Yield: 100%	Utilization of whole cell mediated deracemization in a chemoenzymatic synthesis of enantiomerically enriched polycyclic chromeno[4,3-b] pyrrolidines
1.1 Reagents: Ethanolamine Catalysts: Triphenylphosphine, Palladium, Cuprous iodide Solvents: Water; 30 min, 35 °C		By: Saravanan, Thangavelu; et al
1.2 10 - 24 h, 85 °C		Organic & Biomolecular Chemistry (2014), 12(26), 4682-4690.
Experimental Protocols		

Scheme 28 (1 Reaction)

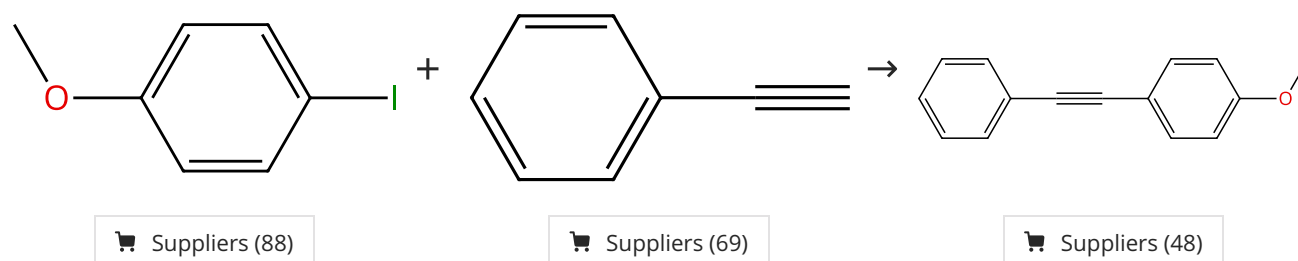
Steps: 1 Yield: 100%



31-180-CAS-8001960	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Potassium carbonate Catalysts: Palladium (silica-supported) Solvents: Ethanol; 4 h, reflux		By: Pandarus, Valerica; et al
Experimental Protocols		Organic Process Research & Development (2012), 16(1), 117-122.

Scheme 29 (1 Reaction)

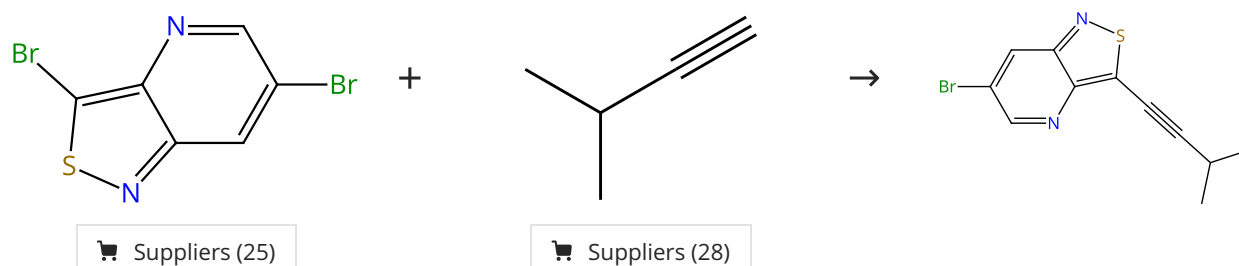
Steps: 1 Yield: 100%



31-180-CAS-2447840	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Potassium carbonate Catalysts: Palladium (silica-supported) Solvents: Ethanol; 4 h, reflux		By: Pandarus, Valerica; et al
Experimental Protocols		Organic Process Research & Development (2012), 16(1), 117-122.

Scheme 30 (1 Reaction)

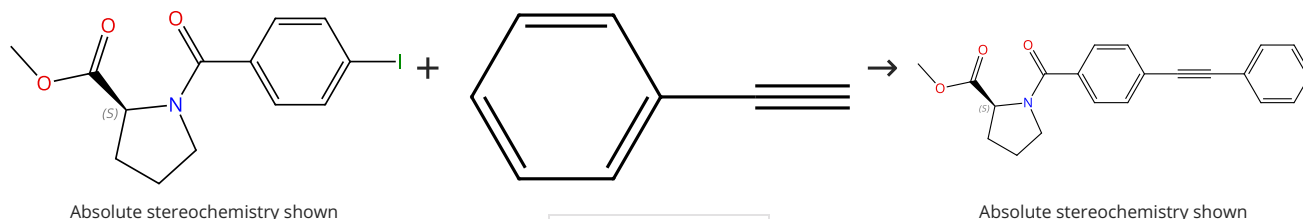
Steps: 1 Yield: 99%



31-180-CAS-19640185 Steps: 1 Yield: 99%	Cyclin G-associated kinase (GAK) affinity and antiviral activity studies of a series of 3-C-substituted isothiazolo[4,3-b]pyridines By: Wouters, Randy; et al European Journal of Medicinal Chemistry (2019), 163, 256-265.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; overnight, rt	
Experimental Protocols	

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

Absolute stereochemistry shown

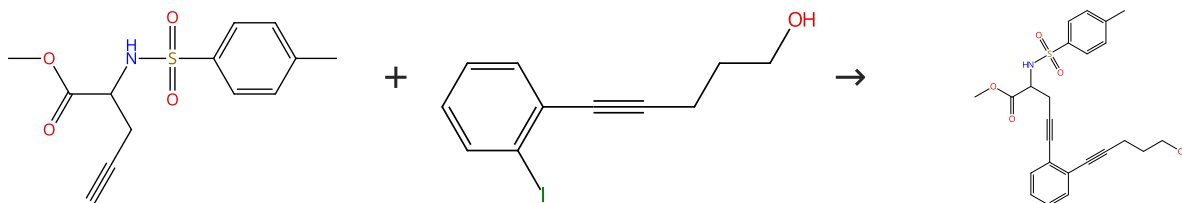
Suppliers (6)

Suppliers (69)

31-180-CAS-16307756 Steps: 1 Yield: 99%	Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography By: Kalinin, Dmitrii V.; et al Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Acetonitrile; 2 h, rt; overnight, rt	
Experimental Protocols	

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



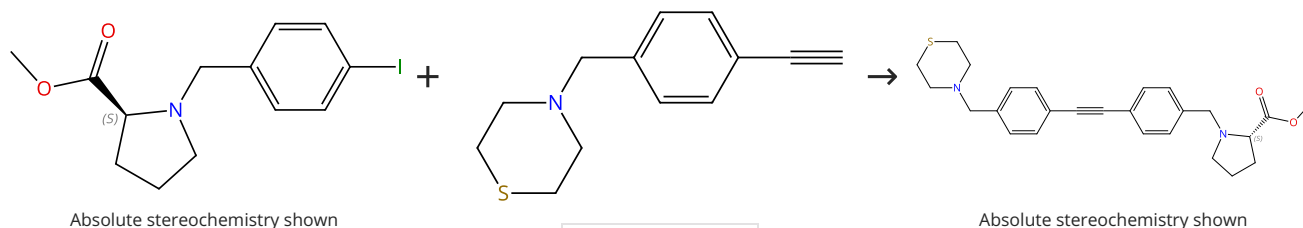
Suppliers (5)

Suppliers (3)

31-180-CAS-2636772 Steps: 1 Yield: 99%	Synthesis and aromatization of cyclic enediyne-containing amino acids By: Kaiser, Jasper; et al Organic & Biomolecular Chemistry (2009), 7(4), 695-705.
1.1 Reagents: Diethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Diethyl ether; overnight, rt	
Experimental Protocols	

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

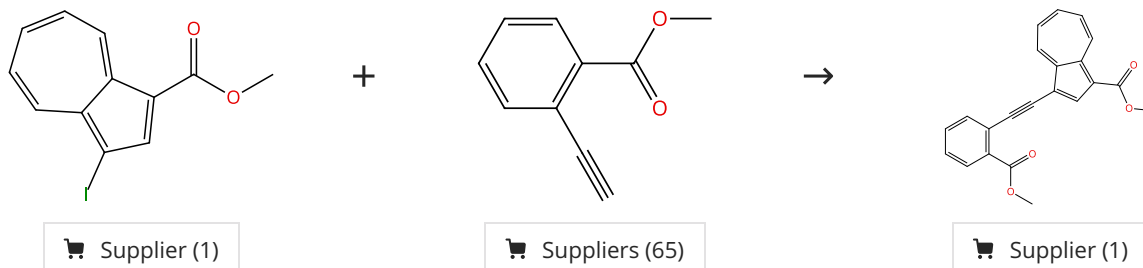
Absolute stereochemistry shown

Supplier (1)

31-180-CAS-16307836	Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Acetonitrile; 2 h, rt; overnight, rt	By: Kalinin, Dmitrii V.; et al Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.
Experimental Protocols	

Scheme 34 (1 Reaction)

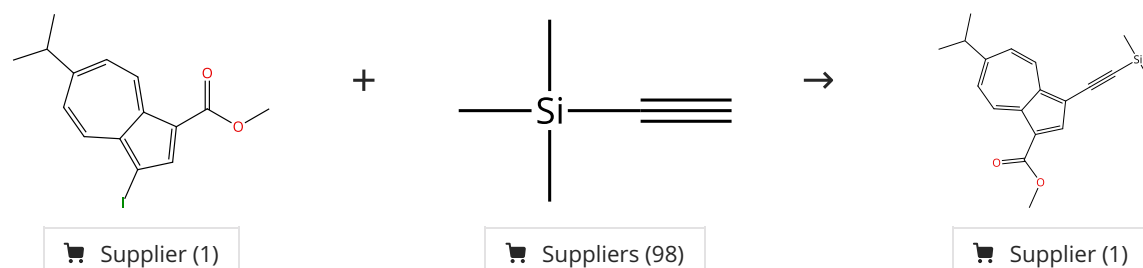
Steps: 1 Yield: 99%



31-180-CAS-18333063	Synthesis of azulene-substituted benzofurans and isocoumarins via intramolecular cyclization of 1-ethynylazulenes, and their structural and optical properties
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 15.5 h, 50 °C	By: Shoji, Taku; et al Organic & Biomolecular Chemistry (2018), 16(3), 480-489.
Experimental Protocols	

Scheme 35 (1 Reaction)

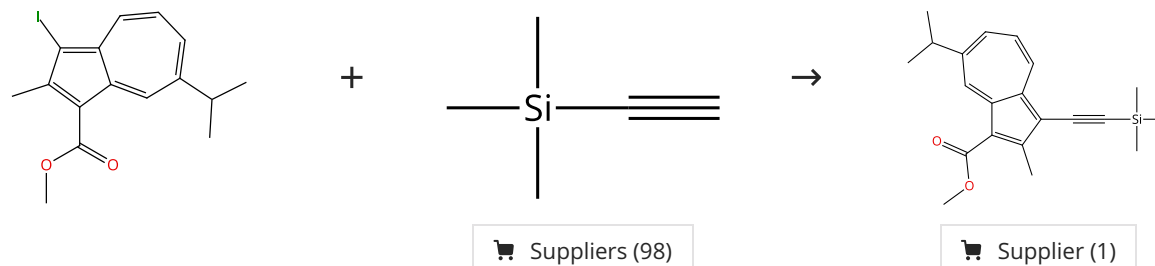
Steps: 1 Yield: 99%



31-180-CAS-18333051	Synthesis of azulene-substituted benzofurans and isocoumarins via intramolecular cyclization of 1-ethynylazulenes, and their structural and optical properties
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 21 h, 50 °C	By: Shoji, Taku; et al Organic & Biomolecular Chemistry (2018), 16(3), 480-489.
Experimental Protocols	

Scheme 36 (1 Reaction)

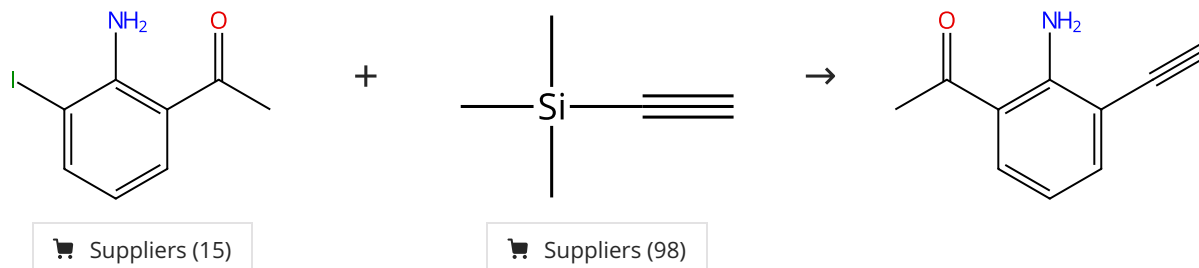
Steps: 1 Yield: 99%



31-180-CAS-18333053	Steps: 1 Yield: 99%	Synthesis of azulene-substituted benzofurans and isocoumarins via intramolecular cyclization of 1-ethynylazulenes, and their structural and optical properties
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 12 h, 50 °C		By: Shoji, Taku; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2018), 16(3), 480-489.

Scheme 37 (1 Reaction)

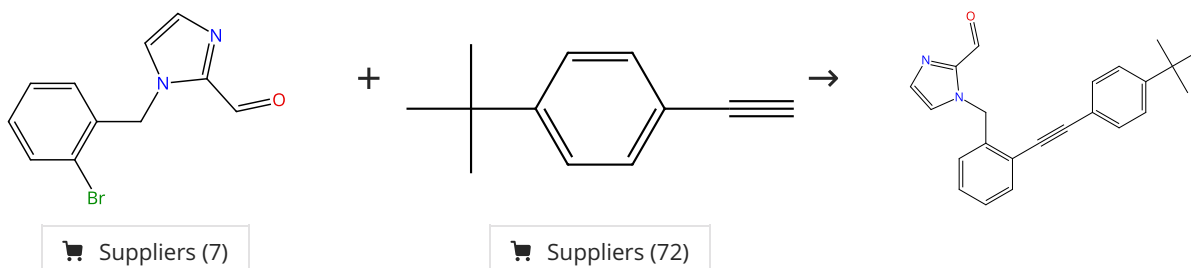
Steps: 1 Yield: 99%



31-180-CAS-7895714	Steps: 1 Yield: 99%	A dramatic enhancing effect of InBr₃ towards the oxidative Sonogashira cross-coupling reaction of 2-ethynylanilines
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; rt; overnight, rt		By: Ikeda, A.; et al
1.2 Reagents: Water; rt		Organic & Biomolecular Chemistry (2016), 14(6), 2127-2133.
1.3 Reagents: Potassium carbonate Solvents: Methanol, Water; rt		
Experimental Protocols		

Scheme 38 (1 Reaction)

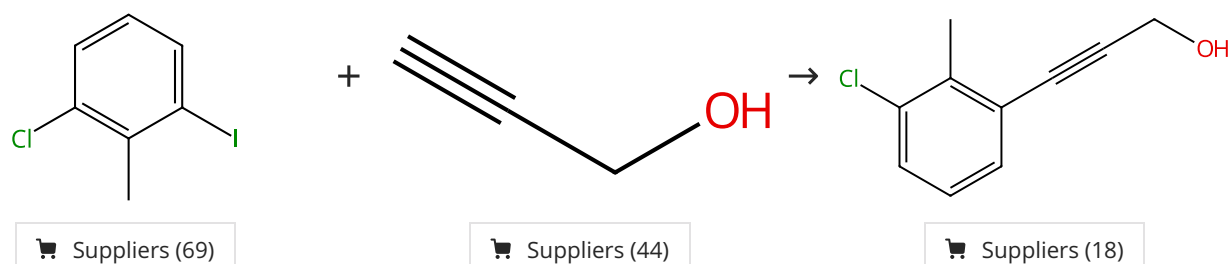
Steps: 1 Yield: 99%



31-614-CAS-34896291	Steps: 1 Yield: 99%	Synthesis of imidazo[1,2-a]benzoazepines by alkyne-carbonyl-metathesis
1.1 Reagents: Diisopropylamine Catalysts: Cuprous iodide, Palladium, bis(acetonitrile)dichloro-, X-Phos Solvents: 1,4-Dioxane; 5 min, 20 °C; 20 °C → 35 °C		By: Sobhani, Maryam; et al
1.2 35 °C; 12 h, 35 °C		Organic & Biomolecular Chemistry (2022), 20(46), 9207-9216.
Experimental Protocols		

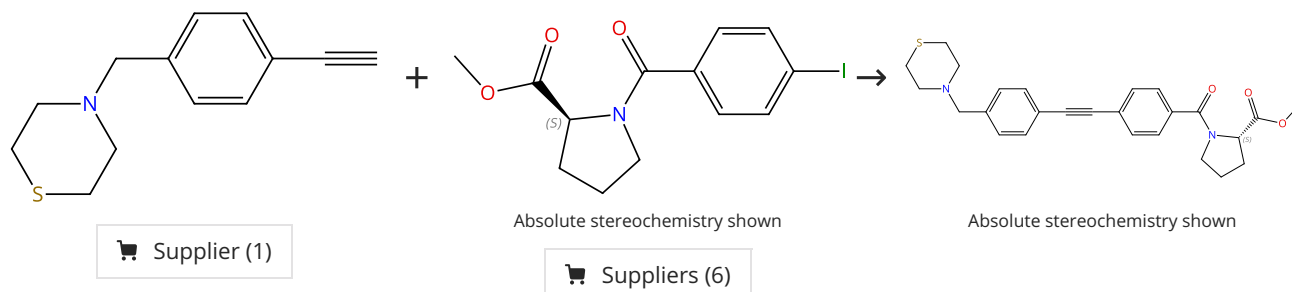
Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



31-180-CAS-6447930	Steps: 1 Yield: 99%	Synthesis of 3-benzylisoquinolines by domino imination/ cycloisomerisation of 2-propargylbenzaldehydes
1.1 Reagents: Diethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium; 15 min, rt		By: Dell'Acqua, Monica; et al
1.2 Catalysts: Cuprous iodide; 3 h, 50 °C		Organic & Biomolecular Chemistry (2014), 12(40), 8019-8030.
Experimental Protocols		

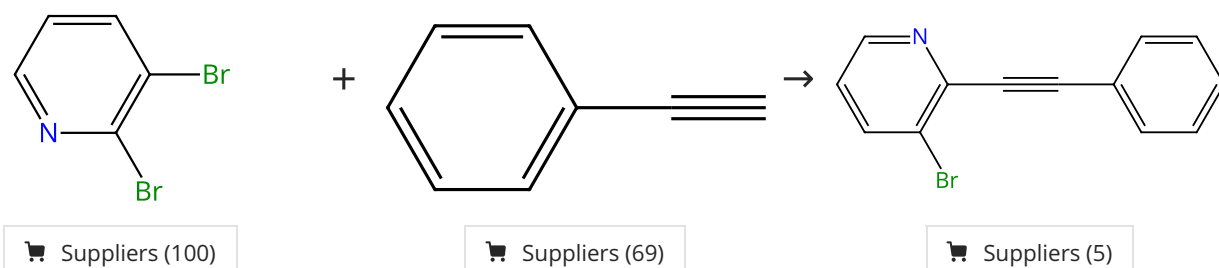
Scheme 40 (1 Reaction)

Steps: **1** Yield: **99%**

31-180-CAS-16307754	Steps: 1 Yield: 99%	Novel Potent Proline-Based Metalloproteinase Inhibitors: Design, (Radio)Synthesis, and First in Vivo Evaluation as Radiotracers for Positron Emission Tomography
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Acetonitrile; 2 h, rt; overnight, rt		By: Kalinin, Dmitrii V.; et al Journal of Medicinal Chemistry (2016), 59(20), 9541-9559.
Experimental Protocols		

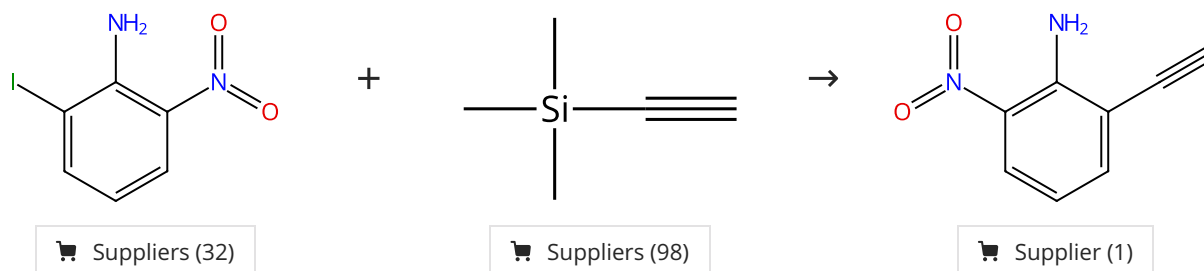
Scheme 41 (1 Reaction)

Steps: **1** Yield: **99%**



31-180-CAS-22409146	Steps: 1 Yield: 99%	Synthesis of thieno[2,3-h]/[3,2-h]quinolines and thieno[2,3-f]quinolines by Bronsted acid mediated cycloisomerisation By: Ponce, Marian Blanco; et al Organic & Biomolecular Chemistry (2020), 18(33), 6531-6536.
1.1	Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine) palladium Solvents: Diisopropylamine; 0.5 h, 40 °C	
Experimental Protocols		

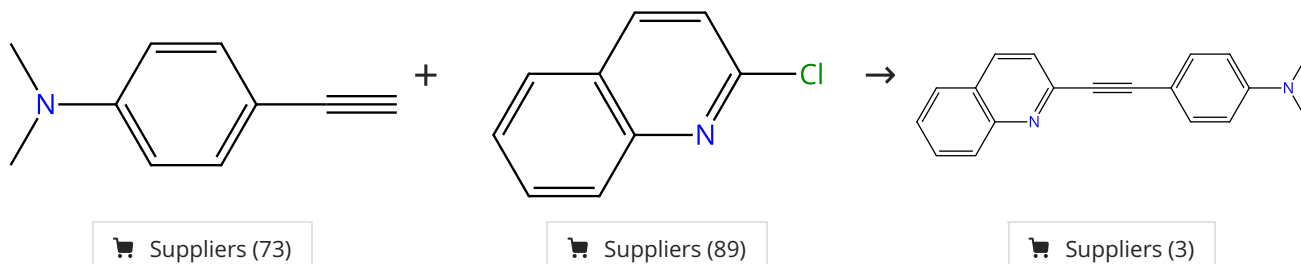
Scheme 42 (1 Reaction)

Steps: **1** Yield: **99%**

31-180-CAS-1397737	Steps: 1 Yield: 99%	A dramatic enhancing effect of InBr₃ towards the oxidative Sonogashira cross-coupling reaction of 2-ethynylanilines By: Ikeda, A.; et al Organic & Biomolecular Chemistry (2016), 14(6), 2127-2133.
1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; rt; overnight, rt		
1.2 Reagents: Water; rt		
1.3 Reagents: Potassium carbonate Solvents: Methanol, Water; rt		
Experimental Protocols		

Scheme 43 (1 Reaction)

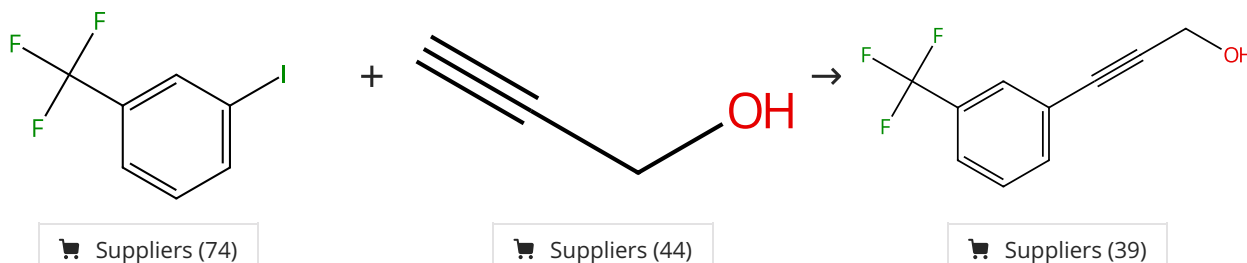
Steps: 1 Yield: 99%



31-180-CAS-2581006	Steps: 1 Yield: 99%	Synthesis and electrogenerated chemiluminescence of donor-substituted phenylquinolinylethynes and phenylisoquinolinylethynes: effect of positional isomerism By: Elangovan, Arumugasamy; et al Organic & Biomolecular Chemistry (2004), 2(11), 1597-1602.
1.1 Reagents: Triethylamine, Hydrogen Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; 80 °C; 24 h, reflux		

Scheme 44 (1 Reaction)

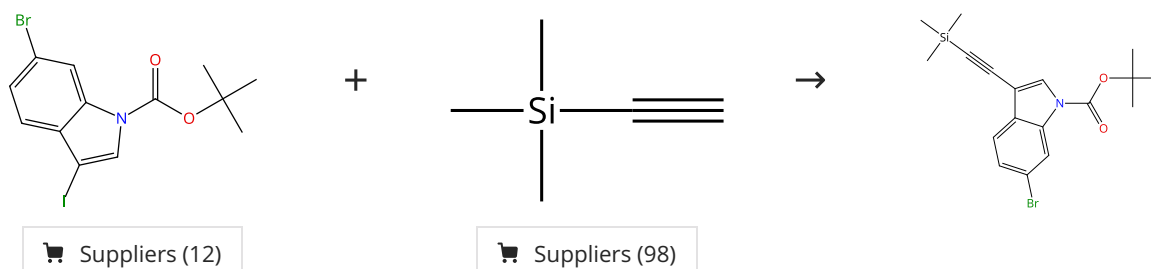
Steps: 1 Yield: 99%



31-180-CAS-8360717	Steps: 1 Yield: 99%	Synthesis of 3-benzylisoquinolines by domino imination/ cycloisomerisation of 2-propargylbenzaldehydes By: Dell'Acqua, Monica; et al Organic & Biomolecular Chemistry (2014), 12(40), 8019-8030.
1.1 Reagents: Diethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium; 15 min, rt		
1.2 Catalysts: Cuprous iodide; 2 h, 50 °C		
Experimental Protocols		

Scheme 45 (1 Reaction)

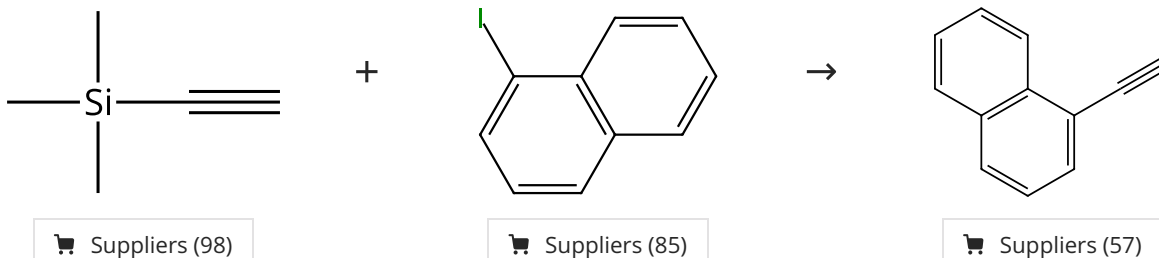
Steps: 1 Yield: 99%



31-180-CAS-17793663 Steps: 1 Yield: 99% 1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Dimethylformamide; 2 h, 60 °C 1.2 Reagents: Water	Synthesis and biological evaluation of the ascidian blood-pigment halocyanine A By: Fong, Hugo K. H.; et al Organic & Biomolecular Chemistry (2017), 15(29), 6194-6204.
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Scheme 46 (1 Reaction)

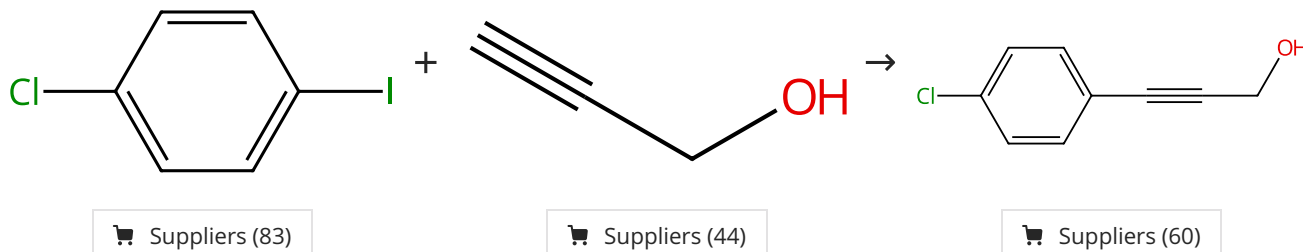
Steps: 1 Yield: 99%



31-180-CAS-5156512 Steps: 1 Yield: 99% 1.1 Reagents: Triethylamine Catalysts: Cuprous iodide, Dichlorobis(triphenylphosphine) palladium Solvents: Tetrahydrofuran; rt; overnight, rt 1.2 Reagents: Potassium carbonate Solvents: Methanol, Water; rt Experimental Protocols	One-pot synthesis of 1,3-enynes with a CF₃ group on the terminal sp² carbon by an oxidative Sonogashira cross-coupling reaction By: Ikeda, Akari; et al Organic & Biomolecular Chemistry (2015), 13(33), 8886-8892.
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Scheme 47 (1 Reaction)

Steps: 1 Yield: 99%



31-180-CAS-4425503 Steps: 1 Yield: 99% 1.1 Reagents: Diethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium; 15 min, rt 1.2 Catalysts: Cuprous iodide; 1.5 h, 50 °C Experimental Protocols	Synthesis of 3-benzylisoquinolines by domino imination/cycloisomerisation of 2-propargylbenzaldehydes By: Dell'Acqua, Monica; et al Organic & Biomolecular Chemistry (2014), 12(40), 8019-8030.
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Scheme 48 (1 Reaction)

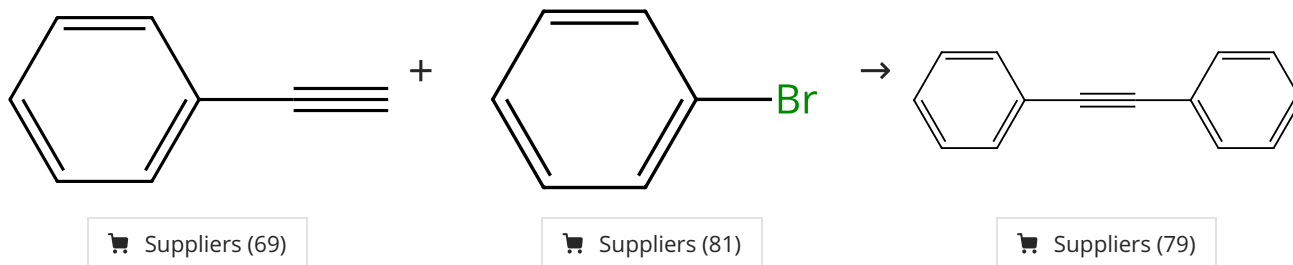
Steps: 1 Yield: 99%



31-180-CAS-12949531 Steps: 1 Yield: 99% 1.1 Reagents: Diethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium; 15 min, rt 1.2 Catalysts: Cuprous iodide; 2 h, 50 °C Experimental Protocols	Synthesis of 3-benzylisoquinolines by domino imination/ cycloisomerisation of 2-propargylbenzaldehydes By: Dell'Acqua, Monica; et al Organic & Biomolecular Chemistry (2014), 12(40), 8019-8030.
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Scheme 49 (1 Reaction)

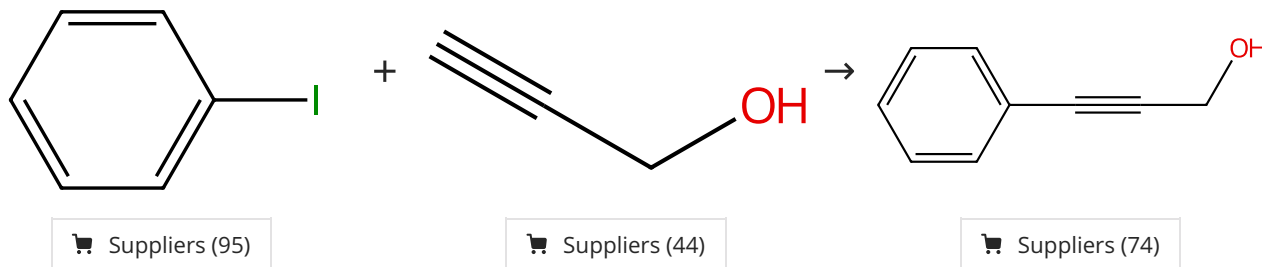
Steps: 1 Yield: 99%



31-180-CAS-10375079 Steps: 1 Yield: 99% 1.1 Reagents: Potassium iodide, Potassium fluoride Catalysts: Palladium Solvents: Dimethylformamide, Water; rt; rt → 130 °C; 48 h, 130 °C	One-pot Pd/C catalyzed domino HALEX and Sonogashira reactions: a ligand- and Cu-free alternative By: Thathagar, Mehul B.; et al Organic & Biomolecular Chemistry (2006), 4(1), 111-115.
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Scheme 50 (1 Reaction)

Steps: 1 Yield: 99%



31-180-CAS-12017272 Steps: 1 Yield: 99% 1.1 Reagents: Diethylamine Catalysts: Dichlorobis(triphenylphosphine)palladium; 15 min, rt 1.2 Catalysts: Cuprous iodide; 1 h, 50 °C Experimental Protocols	Synthesis of 3-benzylisoquinolines by domino imination/ cycloisomerisation of 2-propargylbenzaldehydes By: Dell'Acqua, Monica; et al Organic & Biomolecular Chemistry (2014), 12(40), 8019-8030.
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